NEIOS — N-ATLAS Enterprise Integration OS

Technical Documentation — NAIC 2026, Track: Innovation & Enterprise, Problem Statement PS1 (Developer Infrastructure)
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1. What NEIOS is

NEIOS is governed developer infrastructure for N-ATLAS. It is the layer between a raw model endpoint and an organisational workflow: work is requested through structured intake, authorised by a deterministic policy gate *before* any model call, executed by N-ATLAS, validated against a strict schema, annotated with field-level provenance, routed to human approval when consequential, and recorded in an append-only audit trail. The model never decides its own authority.

Explicitly not: a prompt wrapper, a chat playground, a claim of official N-ATLAS/NITDA/Awarri affiliation, or a system where the LLM decides policy.

2. Architecture

Modular TypeScript monorepo (pnpm). One call path — all model access flows through a single provider interface; all output flows through the same governed loop.

Component	Role
`packages/natlas-adapter`	`NAtlasProvider` interface + OpenAI-compatible implementation. Endpoint, model and key fully configurable via env (`NATLAS_BASE_URL`, `NATLAS_MODEL`, `NATLAS_API_KEY`) — works against Ollama, vLLM or HF Inference Endpoints with zero code changes.
`packages/governance`	The governed loop: APSOS intake packager, deterministic policy engine, invoice schema validation + deterministic arithmetic repair, field provenance annotator, approval routing, JSONL audit writer.
`packages/neios-mcp`	MCP server (stdio) — the governed loop exposed as 5 tools (`neios_submit_request`, `neios_pending_approvals`, `neios_record_decision`, `neios_get_request`, `neios_audit_tail`) with machine-readable `structuredContent`, for Claude Desktop / IDEs / agent frameworks. The tool contract caps at `pending_approval` — no tool can execute consequential work.
`benchmarks/eval.ts`	Golden-fixture evaluation harness — 10 cases across 4 languages + injection + missing-field + arithmetic-trap, scored report pinned to a prompt version (`engineering/metrics/`).
`services/natlas-asr`	Local FastAPI shim serving the four official NCAIR1 ASR checkpoints at the official `/v1/audio/transcriptions` shape — any N-ATLAS app can accept spoken Hausa/Yoruba/Igbo/Nigerian English without its own ASR stack.
`apps/vlthr-hr-web`	**Reference implementation** — Next.js 15 control-first UI + API routes (`/api/work`, `/api/approve`, `/api/audit`). Shows the live 8-stage execution trace — not a chat box.
`data/audit/audit-log.jsonl`	Append-only audit trail; every run — successful, blocked, failed, approved, rejected — appends a record with input/output SHA-256 hashes.

3. The governed loop

```
REQUEST → CONTEXT → POLICY → MODEL → RESULT → EVIDENCE → APPROVAL → AUDIT
```

Stage	Behaviour
REQUEST	Human/system work request enters with a request ID.
CONTEXT	APSOS intake (WHO/WHAT/WHY/WHEN) packaged; missing material fields flagged before any model call.
POLICY	Deterministic gate. Deny-by-default. BLOCK short-circuits the pipeline — the model is never invoked. REVIEW/ALLOW proceed.
MODEL	The only place the provider is called. Latency and model identity recorded.
RESULT	Output parsed into a strict `invoice_draft` schema; arithmetic verified deterministically and repaired where verifiable — each correction is named in the trace and audit record. One bounded retry on malformed JSON.
EVIDENCE	Every field annotated `input_fact` / `model_derived` / `system_derived` — claims vs. verified inputs are distinguishable.
APPROVAL	Lane-2+ (consequential) output remains a draft until a human approves or rejects; the decision itself is audited.
AUDIT	Record appended for every run regardless of outcome — including policy blocks and model failures.

4. Model integration

- **Model:** N-ATLaS 8B — `hf.co/QuantFactory/N-ATLaS-GGUF:Q4_K_M` (ungated community GGUF mirror of the official NCAIR1 weights).
- **Serving:** Ollama on a rented RTX 3090 (vast.ai), OpenAI-compatible API at `/v1/chat/completions`.
- **Access:** SSH tunnel only (`127.0.0.1:18000`). No public model port — Ollama ships without auth, so the auto-mapped port is explicitly not used.
- **Fallback path (documented):** vLLM serving the gated `NCAIR1/N-ATLaS` weights with `HF_TOKEN`; the adapter is endpoint-agnostic.

5. Verified results (evidence: `engineering/evidence/`, `data/audit/`)

Metric	Value
Cold latency (first call after model load)	45.7 s
Warm latency (min / median / max)	1.76 / 3.66 / 4.15 s (6 runs)
Languages verified live	English/Nigerian Pidgin, Hausa, Yoruba, Igbo — real work requests, not translated demos
Audit records written	83 — incl. pending_approval, approved, rejected, policy-blocked and controlled failures
Benchmark (live N-ATLaS)	10/10 fixtures pass, mean score 93/100, mean latency ~4.4 s, prompt `invoice-v1` (`engineering/metrics/eval-invoice-v1-*.json`)

MCP live session	Real stdio session vs live N-ATLaS: Pidgin invoice ₦1,085,000 (subtotal+total repaired), Hausa ₦367,500, wire_transfer BLOCK in 0 ms (no model call), injection held at pending_approval, decision→lookup→audit round trip (`engineering/evidence/mcp-live-session-2026-10-12.txt`)
Failure handling	Endpoint unreachable → MODEL fails, downstream skips, audit still appended. No unhandled errors.
Arithmetic enforcement	Model emitted 20×8000 as 1,600,000 — caught by validator, corrected deterministically to 160,000/2,410,000, repairs named in trace + audit

Live trace — Hausa request → approved invoice (real output)

```
✓ REQUEST   request f1339a9f-… from demo-user
✓ CONTEXT   APSOS intake packaged for role "procurement-assistant"
! POLICY    REVIEW — lane ≥ 2 — human approval required before output is final
✓ MODEL     openai-compat:http://127.0.0.1:18000/v1 responded in 3679ms (N-ATLaS-GGUF:Q4_K_M)
✓ RESULT    schema + arithmetic valid
✓ EVIDENCE  field provenance assigned (input_fact / model_derived / system_derived)
! APPROVAL  draft pending human approval (lane 2)
✓ AUDIT     record appended to data/audit/audit-log.jsonl
→ human approved → linked audit record: finalState "approved", total ₦2,410,000
```

Policy block — the model is never called

```
✓ REQUEST   request … from demo-user
✓ CONTEXT   APSOS intake packaged
✗ POLICY    BLOCK — wire/payment transfer outside role authority
· MODEL/RESULT/EVIDENCE/APPROVAL   skipped (blocked by policy)
✓ AUDIT     record appended — finalState "blocked"
```

6. Security & operations

- Policy before model — the model cannot escalate its own authority (deterministic control, not prompt instruction).
- Secrets in `.env.local` only — gitignored, never committed; history verified clean pre-push.
- SSH-tunnelled model access; no public inference port.
- Append-only audit with SHA-256 input/output hashes — tamper-evident.
- Full deployment + restart runbook: `engineering/runbooks/VASTAI-NATLAS.md`. Running cost to date: ~$4.59 of a $5 credit (itemised in `engineering/COST-LEDGER.md`).

7. Reproduce

```
corepack pnpm install
ssh -N -L 18000:localhost:11434 -p <port> root@<host>   # model tunnel (any N-ATLaS endpoint works)
node --experimental-strip-types scripts/natlas-smoke-test.ts
node --experimental-strip-types scripts/neios-run.ts [--blocked|--missing]
corepack pnpm --filter vlthr-hr-web dev                  # http://localhost:3000
corepack pnpm --filter @neios/mcp test                   # stub suite — governed path, no endpoint needed
corepack pnpm --filter @neios/mcp test:live              # real N-ATLaS via NATLAS_BASE_URL
```

8. Live endpoint status (read before evaluation)

The live runs shown in the demo video and the evidence transcripts were executed against real N-ATLaS on a rented GPU instance (vast.ai), since stopped — the compute budget is exhausted, and no hosted N-ATLAS API exists yet. The system is endpoint-agnostic by design: `NATLAS_BASE_URL` is the only configuration knob. If the organisers provide an N-ATLAS API key or hosted endpoint for evaluation, the full benchmark and MCP scenario suite can be re-run and results returned within a day. Without an endpoint, `pnpm --filter @neios/mcp test` still exercises the entire governed path — policy-before-model, validation, approval lane, audit append — against a stubbed provider; every stage is labelled in the trace, so the difference between stub and live is visible, not claimed.

9. Status & roadmap

Gate G0 (real N-ATLAS inference through the governed loop) passed 2026-10-08; G1–G5 build-out complete: reference app with voice intake, 10/10 benchmark, 4:51 demo video with real MCP session chapter, public repository with curated history. Remaining: external beta sessions (2 testers, protocol ready in `engineering/validations/BETA-TEST-PLAN.md` — stub suite needs no endpoint; live path needs any N-ATLaS endpoint). Demo video: github.com/298althr/neios/releases/tag/v1.0-naic-2026.